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Unit “ 4 : Session “ 8 : SLO “ 2 | Shared vs Distributed Models “ Real-World Application

■ I. Discussion Question

- Application: Google Search (large-scale web search engine).
- Why distributed computing: the web index is petabytes in size “ far beyond any single machine “ so it is sharded across thousands of servers, each searching its own partition in parallel.
- Scalability: more machines are simply added as query volume and index size grow (horizontal scaling).
- Fault tolerance: data and services are replicated, so individual machine failures do not interrupt search results.
- Low response time: partial results from many nodes are aggregated in parallel, returning answers in milliseconds.
- Other valid examples: Hadoop/MapReduce batch analytics, Netflix video streaming (CDN), blockchain networks, weather simulation clusters.